

Policy AI Lab – HLSD Review

High-Level Solution Design | Stage-Gate 1 | 11 June 2026

Document reviewed	HLSD – Policy AI Lab-V1.1 Version 1.1 Dated 20/05/2026
Vendor	ByteDance
Solution Owner	Dr. Ahmed Aldurwiesh
Recommendation	Approved with conditions

1. Assessment Summary and Decision

The Policy AI Lab is a greenfield AI decision-intelligence platform designed to transform MOMAH's policy lifecycle from a fully manual, document-centric process into an AI-augmented, evidence-based capability. The HLSD (v1.1, submitted 20/05/2026, prepared by ByteDance) addresses the HLSD mandatory requirements.

This review incorporates the approved implementation card from the AI strategy which specifies 14 named agents across two families (Project Study Agents and Drafting & Generation Agents).

The HLSD is broadly coherent with the approved scope and demonstrates a strong command of solution design at a conceptual level, with well-developed capability descriptions, a detailed agentic AI architecture, and a comprehensive compliance framework. All mandatory checklist items for Stage-Gate 1 are present. However, the HLSD does not explicitly trace its 13 capabilities to the implementation card's 14 named agents, with a total of 10 conditions identified across the review.

Recommendation: Approved with conditions

Consolidated findings:

Ref	Section	Type	Finding	Section outcome
G-01	3	Clarification	Capabilities do not match needs of all 14 agents in AI Strategy	Approved with condition
G-02	4	Clarification	Requirements do not match needs of all 14 agents in AI Strategy	Approved with condition
G-03	5	Clarification	Integration protocols pending SDD	Approved with condition
G-04	6	Clarification	In and out of scope list non-exhaustive	Approved with condition
G-05	8	Clarification	Section 8.2 Identified Risks Flags, overlaps with section 12.3 Execution Risks, with distinction required for what are actual AI Risks compared to use-case delivery risks	Approved with condition
G-06	9	Clarification	No infrastructure sizing provided	Approved with condition
G-07	11	Outstanding	No costs savings, revenue generation, or productivity enhancements mentioned for strategic KPIs	Rejected
G-08	11	Outstanding	KPIs link to business outcomes missing for Vision 2030 alignment scoring and regulatory escalations count post-launch	Rejected

Ref	Section	Type	Finding	Section outcome
G-09	11	Outstanding	KPI target rationale missing as well as assumptions backing figures	Rejected
G-10	12	Outstanding	Delivery plan of MVP releases missing	Rejected

2. Section-by-Section Review

Executive Summary

The executive summary is well-structured and covers all required dimensions: solution purpose, target users, expected outcomes, success metrics, business case status, AI risk classification, and key design assumptions. The decision-intelligence framing is clear and accessible to a non-technical governance audience.

Section decision: Approved with no gap

Section 1: Introduction

Purpose, scope, intended audience, and definitions are documented. In-scope capability groupings are clearly structured across four pillars. Out-of-scope exclusions are explicit and appropriate.

Section decision: Approved with no gap

Section 2: Business Context

The business problem is specific, evidence-based, and well-articulated. The five core pain points are grounded in the current-state reality of a fully manual process and directly align with the AI Strategy's strategic objectives (100% modernization of legislation, strengthen legislative environment, increase homeownership and citizen satisfaction).

Key stakeholders are documented with roles, responsibilities, names, and contacts. The Solution Owner (Dr. Ahmed Aldurwiesh) and AI Champion (Hamad Alsharida) are correctly identified.

Section decision: Approved with no gap

Section 3: Solution Overview

Personas, capabilities, workflows, and user journeys are clearly defined with personas role and needs well documented. Further, capabilities are clearly defined and linked to personas with before/after workflow diagrams present for every core process.

However, the capabilities described are not mapped to the 14 agents/features present in the AI Strategy. Particularly the following agents are only partially covered:

- Policy Flagging Agent: While feature describes a proactive monitoring function to flag new laws and predict their impact on existing policies in the future, CAPO4 and CAPO2 in the HLSD partially cover this scope through reactive conflict scanning and gap analysis
- Operational Impact Analyzer: While feature describes measuring a policy's effect on operations and regulations to identify compliance adjustments needs, CAPO6 (simulation) and CAP11 (post implementation tracking) partially meet these requirements
- Policy Dependency Locator Agent: While feature includes the analysis of other entities' policies (i.e., cross-ministry), the HLSD focuses on building a repository of existing MOMAH policies only (CAPO1)

Further, the Implementation Risk and Readiness Evaluator, which focuses on predicting execution risk and feasibility, has no matching capability as only CAP11 tracks implementation outcomes only post-implementation rather than at the policy drafting stage.

Gap G01 – Capabilities do not match needs of all 14 agents in AI Strategy

Section decision: Approved with condition – Capabilities to be reassessed to match all needs from AI Strategy

Section 4: Requirements Summary

Business requirements are captured and prioritized based on scope and initiative objectives with non-functional requirements specific and measurable. However, the business requirements do not meet all of the needs of the AI Strategy with Policy Update Flagging Agent, Operational Impact Analyzer, Policy Dependency Locator Agent, and Implementation Risk & Readiness Evaluator features not reflected in existing business requirements.

Gap G02 – Requirements do not match needs of all 14 agents in AI Strategy

Section decision: Approved with condition – Requirements to be reassessed to match all needs from AI Strategy

Section 5: High-Level Architecture

Architecture diagram is present and clear with key components and technologies described. Further, integrations are specified with their system, data direction, and protocol, however most protocols are pending confirmation in SDD.

Gap G03 – Protocols pending SDD

Section decision: Approved with condition – Protocols to be specified in final HLSD

Section 6: AI Approach

AI approach and model selection is justified with rationale and alternatives considered. Further, human oversight is defined for material/regulated outputs.

However, intended use cases are incomplete as they do not match the capabilities or the core AI capabilities. Further, the out-of-scope list should also be further detailed to match the details shown in previous sections

Gap G04 – In and out of scope list non-exhaustive

Section decision: Approved with condition – Provide an exhaustive list for scope to ensure alignment early on

Section 7: Data Landscape

Data residency requirements are considered with all data residing in the Kingdom.

Section decision: Approved with no gap

Section 8: AI Risk Classification and Compliance

Risk classification assigns High Risk with a detailed four-point rationale. Six risk flags are documented with likelihood, impact, and proposed controls. The Risk Escalation Framework defines five threshold categories with tiered alert protocol. The compliance checklist covers ten items.

Gap G05 – Section 8.2 Identified Risks Flags, overlaps with section 12.3 Execution Risks, with distinction required for what are actual AI Risks compared to use-case delivery risks

Section decision: Approved with condition – Removal of duplicated execution risks between section 8.2 and section 12.3

Section 9: Infrastructure and Operations

Infrastructure considerations identify requirements at a conceptual level with sizing deferred to Phase 2.

Gap G06 – No sizing provided

Section decision: Approved with condition – Provide high-level infrastructure sizing linked to expected users, load, and solution scope

Section 10: Assumptions, Constraints and Prerequisites

Key assumptions, constraints, and prerequisites are explicitly stated with constraints linked to requirements, architecture, and timeline.

Section decision: Approved with no gap

Section 11: Success Metrics

Strategic, operational and technical KPIs are provided. However, strategic KPIs are missing key metrics with KPIs not directly traceable to all business outcomes.

The following strategic KPIs are to be considered:

- Cost savings in SAR m
- Revenue generation in SAR m
- Productivity enhancement in SAR m

Further, the rational for targets stated are missing with no assumptions stated. Finally, numeric thresholds for technical KPIs are pending SDD.

Gaps:

- G07 – No costs savings, revenue generation, or productivity enhancements mentioned for strategic KPIs

- G08 – KPIs link to business outcomes missing for Vision 2030 alignment scoring and regulatory escalations count post-launch
- G09 – KPI target rationale missing as well as assumptions backing figures

Section decision: Rejected – Strategic KPIs to define cost savings, revenue generation, or productivity enhancements to justify design of solution, KPI list to be expanded to ensure linkage to all business outcomes, and target rationale with assumptions to be clearly stated where possible

Section 12: Roadmap and Phasing

The delivery roadmap needs to be updated to reflect the delivery plan of expected MVP releases (i.e., prototype, alpha form, beta form etc.) as current HLSD specifies that the solution will be rolled out in its complete form directly.

G10 – Delivery plan of MVP releases missing

Section decision: Rejected – Delivery plan of MVP releases to be included

Appendix

The sign-off table in Appendix C contains all required stakeholder rows with signature and date fields to be completed